

Neha R

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TECHNICAL SKILLS

- **AI & ML** : Python, LLMs, NLP (NLU/NLG), Computer Vision
- **GenAI & RAG** : LangGraph (State-Machine), LangChain, Multi-Agent Orchestration
- **Data & Ops** : Pinecone Vector DB, PostgreSQL, Docker, Git, CI/CD Pipelines
- **Development** : FastAPI, Pydantic Validation, Transformer OCR, OpenCV
- **Cloud** : AWS (Certified Solutions Architect – Associate)

PROJECTS

Agentic AI eBook Assistant — RAG System with LLM Evaluation 🔒

Feb 2026

LangGraph, Python, FastAPI, Pinecone, Pydantic, Gemini 2.5 Flash

- Designed a multi-node agentic RAG pipeline (Retriever → Relevance Checker → Generator → Validator) using LangGraph state machines, reducing out-of-context LLM responses from ~30% to under 5% on a structured 50-query evaluation benchmark.
- Applied statistical evaluation methodology: measured retrieval precision, response latency (P50/P95), and hallucination rate across the benchmark set — the same eval-rooted-in-real-scenarios approach used in production ML systems.
- Enforced Pydantic structured output schemas at every node transition and embedded a compile-time retry counter in shared graph state to eliminate infinite execution loops — a failure-mode-first engineering approach.
- Validated pipeline behaviour using pytest — functional tests for in-scope queries, negative tests for out-of-scope inputs, and performance assertions verifying P95 latency under 2s across the 50-query benchmark set.

Loan Buddy — Multi-Agent AI System with On-Device LLM Inference 🔒

Dec 2025

Python, Model Quantization (4-bit/8-bit), Multi-Agent Architecture, FastAPI, Docker

- Architected a five-agent offline AI system (Router, Sales, KYC, Underwriting, Document Generator) coordinated by a Master Orchestrator pattern — automating the full loan lifecycle from conversational intake to sanction letter generation.
- Applied PyTorch-based 4-bit and 8-bit post-training quantization using bitsandbytes to compress LLM weights for consumer hardware deployment, reducing memory footprint by 60% with no external API dependencies.
- Implemented tiered rule-based Underwriting Engine with statistical thresholds on debt-to-income ratio and credit score bands — combining ML inference with deterministic business logic for compliance-ready decisions.
- Full system containerized via Docker Compose with FastAPI backend and Streamlit frontend — production-deployable with one command.

AutoForm AI — Computer Vision & NLP Document Processing Pipeline 🔒

Nov 2025

Python, OpenCV, TrOCR (Transformer OCR), spaCy NLP, FastAPI, Docker, JavaScript

- Built an end-to-end offline Computer Vision pipeline: OpenCV performs adaptive thresholding and contour-based line segmentation on document images, feeding individual text-line crops into Microsoft's TrOCR transformer model — the same architecture as BERT/ViT encoder-decoder stacks used in production CV systems.
- Achieved 78% reduction in processing latency (14 min → 3 min) by replacing full-image OCR with segmented-region inference and switching to batched spaCy NLP processing across all extracted fields simultaneously.
- Implemented an intelligent NLP field classifier using spaCy Named Entity Recognition and regex pattern matching to automatically tag extracted text as Name, Phone, Email, or ID — zero manual labelling required.
- Deployed the ML backend as a containerised Docker service with a FastAPI async REST interface, maintaining memory footprint under 2GB for edge-environment compatibility.

EXPERIENCE

Salesforce Intern – Impactive (An Infogain Company)

Oct 2025 – Dec 2025

- Built 8+ Screen and Record-Triggered Automation Flows across CRM modules, reducing manual operational steps by 20–30% — hands-on exposure to enterprise-scale data workflows and client delivery.
- Optimized SOQL query logic reducing report generation from 14 minutes to under 3 minutes; documented architectural decisions for large-scale enterprise system reviews.

EDUCATION

Vellore Institute of Technology (VIT), Bhopal

Jun 2022 – May 2027

Integrated M.Tech in Artificial Intelligence & Machine Learning
CGPA: 8.69/10.0

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** (2026)
- **Salesforce Developer with Agentblazer Champion** (2025)
- **Cloud Computing – NPTEL** (2024)
- **Applied Machine Learning in Python – University of Michigan** (2023)